Linear systems of three unknowns

1 Determine whether each ordered triple is a solution of the corresponding system of equations:

a Triple: $(-9, -7, 4)$
System:
$$-2x + 5y - 3z = -29$$
$$-3x + y + 4z = 38$$
$$-4x + 2y + 3z = 34$$

b Triple: $(1, -5, -3)$
System:
$$4x - 5y + 3z = 20$$
$$4x - 5y - z = 32$$
$$-x + 2y - 5z = 4$$

c Triple: $\left(\frac{3}{4}, \frac{4}{5}, \frac{2}{5}\right)$
System:
$$5x + y + 3z = \frac{23}{4}$$
$$4x + 3y + 5z = \frac{37}{5}$$
$$2x - 4y + 3z = -\frac{1}{2}$$

d Triple: $(1.1, 0.3, 1.7)$
System:
$$3x + y - 5z = -5.9$$
$$5x + y - 4z = -1$$
$$5x - 2y - 4z = -1.9$$

e Triple: $(-8, -1, 2)$
System:
$$x + 5y + 4z = -8$$
$$4x - y + 3z = -25$$
$$2y - z = -4$$

f Triple: $(-6, 3, -9)$
System:
$$x + 5y + 3z = -18$$
$$5x + y - 3z = -3$$
$$5x + 3y - z = -12$$

2 Solve the following systems of equations:

a
$$x - 5y + z = -16$$
$$2x + 5y + 2z = 13$$
$$-x - 5y - 3z = -10$$

b
$$x + y = 6$$
$$y + z = 13$$
$$x + z = 9$$

c
$$x + 3y = -11$$
$$5y + 4z = -33$$
$$x + 2z = 0$$

d
$$5x + 2y + 4z = -13$$
$$6y + 4z = 22$$
$$5x - 2y + 3z = -31$$

e
$$3x + 2y - z = -9$$
$$6x - 4y + z = -20$$
$$3x + 6y + 4z = 5$$

f
$$3x + 3y - z = 6$$
$$2x - 2y + z = 3$$
$$2x + 5y - z = 2$$

g
$$3x + 2y - 2z = 4$$
$$y + 2z = 5$$
$$3x - y - 2z = 7$$

h
$$x = 8(y - z)$$
$$2z = 9(y - x)$$
$$x + z = 2y - 3$$

i
$$x = 2(y - z)$$
$$y = 4(z - x)$$
$$x + y = z + 6$$

j
$$9x + 5y - 2z = 25$$
$$x + 4y - 7z = 30$$
$$3x - 6y + 8z = -33$$

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3 How many solutions do each of the following systems of equations have?

a
$$\begin{aligned} 3x + 4y + 2z &= -1 \\ 5x - 2y - 3z &= -23 \\ -3x - 4y - 2z &= 1 \end{aligned}$$

b
$$\begin{aligned} 6x + 7y - 4z &= -13 \\ 12x + 14y - 8z &= -24 \\ 3x - 2y + 5z &= 17 \end{aligned}$$

4 Consider the system:

Equation 1	Equation 2
$2x - 3y - z = -19$	$y - z = -1$

Let $z = t$ be arbitrary.

a Solve Equation 2 for y with respect to t.

b Using your result from part a, solve Equation 1 for x with respect to t.

c Hence, write down the solution to the system with respect to the arbitrary variable t.

5 Consider the system:

Equation 1	Equation 2
$4x + 5y - z = 11$	$x + y + 2z = 9$

Let $z = t$ be arbitrary.

a Solve Equation 2 for y with respect to x and t.

b Solve Equation 1 for x with respect to t.

c Hence, solve for y with respect to t.

d Hence, write down the solution to the system with respect to the arbitrary variable t.

6 Consider the system:

Equation 1	Equation 2
$x - 3y + 3z = 6$	$x - 2y + 4z = 5$

Let $z = t$ be arbitrary.

a Solve Equation 1 for x with respect to y and t.

b Solve Equation 2 for y with respect to t.

c Hence, solve for x with respect to t.

d Hence, write down the solution to the system with respect to the arbitrary variable t.

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7 Consider the following system of three equations:

$$\begin{aligned} 3x + 5y + 8z &= 136 \\ 27x + 45y + 72z &= 872 \\ x + y + z &= 25 \end{aligned}$$

a Does the system have one solution, infinite solution or no solution?

b Is the system consistent or incosistent?

8 Consider the following system of 3 equations:

$$\begin{aligned} 8x + 5y + 8z &= 97 \\ 6x + y + 4z &= 45 \\ x + y + z &= 14 \end{aligned}$$

a Solve the system.

b Is the system independent or dependent?

9 Consider the following system of 3 equations:

$$\begin{aligned} 7x + 3y + 3z &= 3.2 \\ 8x + 5y + 3z &= 4.4 \\ 4x + 9y + 4z &= 5.7 \end{aligned}$$

a Solve the system.

b Is the system consistent or incosistent?

10 Find a system of three linear equations in three variables that has $(-4, -2, -1)$ as a solution.

11 Solve the following systems of equations:

a
$$\begin{aligned} \frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= 13 \\ \frac{1}{x} + \frac{1}{y} &= 9 \\ \frac{1}{x} - \frac{1}{z} &= 2 \end{aligned}$$

b
$$\begin{aligned} \frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= 11 \\ \frac{1}{y} + \frac{1}{z} &= 8 \\ \frac{1}{x} - \frac{1}{y} &= -3 \end{aligned}$$

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12 For each of the following system of 3 equations:

i Solve the system.

ii Determine whether the system is consistent or inconsistent.

iii Determine whether the system is dependent or independent.

a
$$\begin{aligned} x - 5z &= -8 \\ y + 4z &= 16 \\ 6y + 2z &= 30 \end{aligned}$$

b
$$\begin{aligned} x + 3z &= 22 \\ y + 8z &= 49 \\ 8z &= 48 \end{aligned}$$

c
$$\begin{aligned} x + 5y - 7z &= -8 \\ y + 6z &= 56 \\ 6y + 9z &= 120 \end{aligned}$$

d
$$\begin{aligned} 9x + 9y + 6z &= 303 \\ 16x + 18y + 10z &= 536 \\ x + y + z &= 39 \end{aligned}$$

e
$$\begin{aligned} 5x + 4z &= 42 \\ 5x + 9y + &= 37 \\ 6y + 7z &= 74 \end{aligned}$$

f
$$\begin{aligned} 9x + 7y &= 180 \\ x + y + 5z &= 27 \\ 4x + 6y + 4z &= 110 \end{aligned}$$

g
$$\begin{aligned} 9x + 7y + 3z &= 151 \\ 8x + y + 9z &= 142 \\ x + y + z &= 23 \end{aligned}$$

h
$$\begin{aligned} 8x + 9y + 2z &= 51 \\ 4x + 8y + 9z &= 101 \\ 8x + 4y + 9z &= 109 \end{aligned}$$

i
$$\begin{aligned} 3x + 7y &= 4.4 \\ x + y + 6z &= 3.8 \\ 5x + 6y + 8z &= 8.5 \end{aligned}$$

Applications

13 Consider the equation: $Ax + By + Cz = 12$.

a One solution to this equation is $\left(1, \frac{3}{4}, 3\right)$. Substitute these values into the equation, to give an equation in terms of A, B, and C.

b A second solution to the equation is $\left(\frac{4}{3}, 1, 2\right)$. Substitute these values into the equation, to give an equation in terms of A, B, and C.

c A third solution to the equation is $(2, 1, 1)$. Substitute these values into the equation, to give an equation in terms of A, B, and C.

d Solve your three equations for A, B, and C.

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14 The fraction $\frac{1}{24}$ can be written as the following sum:

$$\frac{1}{24} = \frac{x}{8} + \frac{y}{4} + \frac{z}{3}$$

where the numbers x, y, and z are solutions of:

$$\begin{aligned} x + y + z &= 1 \quad ------ \text{ equation 1} \\ 2x - y + z &= 0 \quad ------ \text{ equation 2} \\ -x + 2y + 2z &= -1 \quad ------ \text{ equation 3} \end{aligned}$$

a Solve this system of equations for x, y, and z.

b Use your values of x, y, and z to determine whether the following equation is true or false:

$$\frac{1}{24} = \frac{x}{8} + \frac{y}{4} + \frac{z}{3}$$

15 Consider the quadratic with equation $y = ax^2 + bx + c$ that passes through the points $(1, 15)$, $(2, 5)$ and $(4, -3)$.

a Set up a system of equations in the variables a, b and c using the points given.

b Solve your equations for a, b, and c.

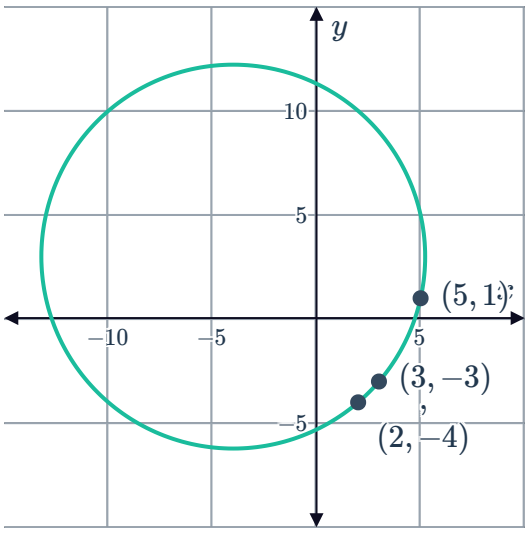
c Write down the quadratic equation that passes through the points $(1, 15)$, $(2, 5)$ and $(4, -3)$.

16 The equation of a circle can be written in the general form $x^2 + y^2 + ax + by + c = 0$. The circle below passes through the points $(2, -4)$, $(5, 1)$ and $(3, -3)$:

a The general form of a circle given may be rearranged to the following form: $ax + by + c = -\left(x^2 + y^2\right)$. Use this form to set up a system of equations in the variables a, b and c from the given points.

b Solve your equations for a, b, and c.

c Write down the equation of the circle that passes through the points $(2, -4)$, $(5, 1)$ and $(3, -3)$.



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